Ahmed **Cheikh Rouhou**

Engineering Manager · PhD in Computer Science

Ariana - Tunisia

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Summary

Engineering Manager leading an AI team of up to 20 through senior technical leads, with a PhD in Computer Science and 10+ years across applied research and production machine learning. Delivered computer vision, NLP and computational pathology systems from research to production, with QA validation for GxP-regulated environments. Hands-on background in Transformers, graph neural networks and MLOps, and author of 10+ peer-reviewed publications. Builds teams and talent through hiring, internship programs and university collaborations, and uses LLM agents and MCP tooling to streamline engineering workflows.

Skills

Leadership	Team and team-lead management, technical hiring, internship programs, sprint planning, SOWs, GxP validation
AI & ML	PyTorch, TensorFlow, Transformers, Computer Vision, NLP, Graph Neural Networks, Foundation Models, AlphaFold, Hidden Markov Models (HTK, Kaldi)
LLM Agents	Claude, AI agent workflows, MCP servers and agent skills for integrating agents into team workflows
MLOps & Cloud	Docker, Kubernetes, cloud platforms, CI/CD, MLflow, Weights & Biases, ONNX
Coding	Python, C/C++, Java/JEE, SQL, Bash
Databases	PostgreSQL, MySQL, Oracle, HDF
Languages	Arabic, French and English

Experiences

InstaDeep Ltd — *Tunis, Tunisia*

Engineering Manager — *Jan. 2026 - Present*

- Lead a team of up to 20 people, including interns, organised around senior technical leads.
- Own quality assurance and validation of AI projects for GxP-regulated environments.
- Define hiring plans and co-create statements of work (SOWs) for projects.

InstaDeep Ltd — *Tunis, Tunisia*

Senior Research Engineer — *Sep. 2023 - Dec. 2025*

- Project lead for a histology (H&E) whole-slide image segmentation system detecting tissue regions: owned the software architecture and sprint planning while remaining a hands-on contributor implementing core tasks.
- Integrated **LongViT** as the backbone, enabling end-to-end processing of whole-slide images at up to 32,768 × 32,768 pixels (about 1M patches) and improving on the baseline; went on to develop and release successive production models, deployed in a separate environment through ONNX. Project ongoing.
- Managed three team leads and, through them, a team of 11; involved in hiring and technical decisions across other projects.
- Owned the internship program end to end, from defining research subjects to interviewing, hiring, follow-up and validation; recruited 12+ interns.
- Led university collaborations with 5+ Tunisian universities, mostly engineering schools, teaching courses on Transformers, LLMs and computer vision; created and organised hackathons, led participant teams in workshops and hackathons, and served as a judge at events including Deep Learning Indaba and IndabaX.

InstaDeep Ltd — *Tunis, Tunisia*

Research Engineer & Team Lead — *Sep. 2021 - Sep. 2023*

- Led and managed a team of 4 (full-time engineers and interns), providing technical direction, day-to-day management and mentoring.
- Researched AI-enabled peptide design and modification in an R&D collaboration with a biotech partner, combining Transformers, graph neural networks and AlphaFold; focused on modelling and predicting peptide binding.
- Co-supervised a PhD thesis with CRNS on continual learning for handwritten-document foundation models; the work produced MSdocTr-Lite (*Pattern Recognition Letters*, 2023) and UCL-MHTR (*Expert Systems with Applications*, 2025).
- Led an internship research project on end-to-end information extraction from handwritten documents: a Transformer + Graph Convolutional Network model (Sparse Graph Transformer Encoder with a Cross-GCN decoder) that jointly performs handwriting and named entity recognition, reaching state-of-the-art results on IAM and the ICDAR 2017 Information Extraction competition. Published at WACV 2024.
- Actively involved in technical hiring for AI roles as a technical leader.

InstaDeep Ltd — *Tunis, Tunisia*

Research Engineer — *Jan. 2020 - Sep. 2021*

- Designed and implemented (PyTorch) an end-to-end Transformer model that jointly performs paragraph-level handwriting recognition and named entity recognition on historical documents: the first Transformer-based approach for NER in handwritten documents, reaching a new state of the art on the ICDAR 2017 Information Extraction competition (complete task). Published in *Pattern Recognition Letters* (2022), in collaboration with the Digital Research Centre of Sfax (CRNS).
- Contributed to the pre-training and data preparation of **TunBERT**, the first open-source BERT model for the Tunisian dialect, built with NVIDIA NeMo/PyTorch on multi-GPU infrastructure in a joint project with NVIDIA and iCompass.
- Released the code and fine-tuning datasets publicly (GitHub) and co-authored two publications on the work (ISPR 2022, SN Computer Science 2023).
- Contributed to the preparation of InstaDeep's NVIDIA GTC talk presenting TunBERT.

Esprit School of Business — *Ariana, Tunisia*

Contractual Teacher — *Sep. 2021 - 2023*

- Taught Machine Learning Fundamentals and Deep Learning Fundamentals.

Digital Research Centre of Sfax (CRNS) — *Sfax, Tunisia*

Researcher & PhD Student — *Mar. 2015 - Dec. 2019*

- Arabic text recognition: word spotting and indexing of Arabic documents.

University of Sfax & Training Pro Group — *Sfax, Tunisia*

Contractual Teacher & Java Instructor — *Sep. 2017 - Sep. 2019*

- University of Sfax: Computer Sciences and Internet Certificate (Faculty of Economics and Management); JEE application tutorial and introduction to neural networks for image classification (National School of Engineers of Sfax).
- Training Pro Group: instructor for Spring Framework, Hibernate ORM, Perl and Python.

Logicom Company — *Sfax, Tunisia*

Full Stack Developer — *Sep. 2013 - Jan. 2016*

- Built a real-time synchronization system (RMI, FTP, SQL) and a dedicated backup module (services, compression, logging).
- Upgraded the software to .NET (C#, WPF, NHibernate), optimized existing code (PL/SQL integration, design patterns, threads) and added device communication (time-attendance controller, cash drawer, RFID, SMS).

Talan Tunisie Internationale — *Tunis, Tunisia*

Full Stack Engineer — *Feb. 2013 - Jul. 2013*

- Created an API for a geographic information system and a web application for marking construction sites in Tunisia.

Education

ENIS (National School of Engineers of Sfax) — *Sfax, Tunisia*

PhD in Computer Science — *2015 - Jun. 2021*

- Thesis: *Keyword Spotting in Printed and Handwritten Multi-Script Document Images*. Defended June 2021 with honors (*Mention Honorable*).
- Supervisors: Yousri Kessentini, Slim Kanoun.

FST (Faculty of Sciences of Tunis) — *Tunis, Tunisia*

Computer Science Engineer — *Sep. 2010 - Aug. 2013*

IPEIS (Sfax Preparatory Engineering Institute) — *Sfax, Tunisia*

Preparatory Classes for Engineering (Math/Physics) — *Sep. 2008 - Aug. 2010*

Hedi Soussi School — *Sfax, Tunisia*

Computer Science Baccalaureate, with honors — *Sep. 2007 - Jun. 2008*

Community & Committees

Ongoing	**Judge, Mentor & Technical Advisor**, AI hackathons, workshops and community events, including Deep Learning Indaba and IndabaX (3+ per year)	*Tunisia*
2023	**Guest Lecturer**, Transformer-Based NLP course, TICV Master's program, National Engineering School of Tunis (ENIT)	*Tunisia*
2023	**Publicity Chair**, International Conference on Intelligent Systems and Pattern Recognition	*MENA*
2020	**AI Trainer**, DSC Academy	*MENA*
2020	**Speaker**, DSC SupCom - Data Science	*Tunis-Tunisia*
2019	**Member of Organization Committee**, Deep Learning INDABAX-Sfax	*Sfax-Tunisia*
2019	**Teacher Assistant**, Deep Learning Spring School (Nvidia)	*Monastir-Tunisie*
2018	**Member of Organization Committee**, 2nd BioDialog Summer School on Biodiversity Informatics	*Hammamet-Tunisia*

Certifications

NVIDIA — *Deep Learning Institute*

NVIDIA Deep Learning Institute: University Ambassador — *2019 - 2025*

- Efficient Large Language Model (LLM) Customization (Jan. 2025)
- Deploying RAG Pipelines for Production at Scale (Jan. 2025)
- Generative AI with Diffusion Models (Dec. 2023)
- University Ambassador Certification (Feb. 2022)
- Building Transformer-Based Natural Language Processing Applications (Jan. 2022)
- Fundamentals of Deep Learning, for Multi-GPUs (Jan. 2022), for Multiple Data Types (Mar. 2019), for Computer Vision
- Model Parallelism: Building and Deploying Large Neural Networks

Udacity — *Nanodegree Program*

Machine Learning DevOps Engineer — *Jan. 2022 - May 2022*

InstaDeep — *Internal*

Leadership Development Programs (2)

Publications

UCL-MHTR: A unified continual learning system for multilingual handwritten text recognition — *Expert Systems with Applications*

Marwa Dhiaf, Mohamed Ali Souibgui, Yousri Kessentini, Alicia Fornés, Ahmed Cheikh Rouhou — *2025*

[Link]

Graph Neural Networks for End-to-End Information Extraction From Handwritten Documents — *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*

Yessine Khanfir, Marwa Dhiaf, Emna Ghodhbani, Ahmed Cheikh Rouhou, Yousri Kessentini — *2024*

[Link]

MSdocTr-Lite: A lite transformer for full page multi-script handwriting recognition — *Pattern Recognition Letters*

Marwa Dhiaf, Ahmed Cheikh Rouhou, Yousri Kessentini, Sinda Ben Salem — *2023*

[Link]

TunBERT: Pretraining BERT for Tunisian Dialect Understanding — *SN Computer Science*

Hatem Haddad, Ahmed Cheikh Rouhou, Abir Messaoudi, Abir Korched, Chayma Fourati, Amel Sellami, Moez Ben HajHmida, Faten Ghriss — *2023*

[Link]

TunBERT: Pretrained Contextualized Text Representation for Tunisian Dialect — *International Conference on Intelligent Systems and Pattern Recognition (ISPR)*

Abir Messaoudi, Ahmed Cheikh Rouhou, Hatem Haddad, Nourchene Ferchichi, Moez BenHajhmida, Abir Korched, Malek Naski, Faten Ghriss, Amine Kerkeni — *2022*

[Link]

Transformer-Based Approach for Joint Handwriting and Named Entity Recognition in Historical documents — *Pattern Recognition Letters*

Ahmed Cheikh Rouhou, Marwa Dhiaf, Yousri Kessentini, Sinda Ben Salem — *2022*

[Link]

Multi-task learning for simultaneous script identification and keyword spotting in document images — *Pattern Recognition*

Ahmed Cheikh Rouhou, Yousri Kessentini, Slim Kanoun — *2021*

[Link]

Hybrid HMM/BLSTM system for multi-script keyword spotting in printed and handwritten documents with identification stage — *Neural Computing and Applications*

Ahmed Cheikh Rouhou, Yousri Kessentini, Slim Kanoun — *2020*

[Link]

Hybrid HMM/DNN System for Arabic Handwriting Keyword Spotting — *International Conference on Image Analysis and Recognition (ICIAR)*

Ahmed Cheikh Rouhou, Yousri Kessentini, Slim Kanoun — *2019*

[Link]

HMM based keyword spotting system in printed/handwritten Arabic/Latin documents with identification stage

International Conference on Image Analysis and Recognition (ICIAR)

Ahmed Cheikh Rouhou, Yousri Kessentini, Slim Kanoun

2019

[Link]

A HMM-Based Arabic/Latin Handwritten/Printed Identification System

International Conference on Hybrid Intelligent Systems (HIS)

Ahmed Cheikh Rouhou, Zeineb Abdelhedi, Yousri Kessentini

2016

[Link]

ALTID : Arabic/Latin Text Images Database for recognition research

International Conference on Document Analysis and Recognition (ICDAR)

Imen Chtourou, Ahmed Cheikh Rouhou, Faten Kallel Jaiem, Slim Kanoun

2015

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